Detecting Anomalous DNS Activity using Zeek, TryHackMe

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Objective

The purpose of this analysis is to investigate a triggered alert titled 'Anomalous DNS Activity'. Using Zeek, the task is to inspect a provided PCAP file (dns-tunneling.pcap) and confirm whether the alert is a true positive. Indicators of DNS tunneling and anomalous traffic will be identified using Zeek logs and filters.

Tools & Environment

- Zeek (Network Security Monitor)
- Ubuntu Virtual Machine
- TryHackMe (Anomalous DNS lab environment)
- Command-line utilities: cat, zeekcut, sort, uniq, head

Scenario Recap

A DNS alert was triggered suggesting possible DNS tunneling — a technique often used by attackers to exfiltrate data via DNS queries. The PCAP file dns-tunneling.pcap was provided for deep packet analysis.

Path to PCAP:
cd Desktop/Exercise-Files/anomalous-dns

Zeek command used:
zeek -C -r dns-tunneling.pcap

Methodology & Commands Used

STEP	DESCRIPTION	COMMAND
1	Inspect DNS log	cat dns.log

2	Preview first 10 lines	cat dns.log | head -n 10
3	Count DNS query types	cat dns.log | zeekcut qtype_name | sort | uniq -c
4	Inspect conn.log	cat conn.log
5	Find longest connection	cat conn.log | zeek-cut duration | sort | uniq
6	Identify involved hosts	cat conn.log | zeek-cut id.orig_h | sort | uniq

Key Findings

1. High Volume of DNS AAAA Queries
- Found: 320 records of AAAA (IPv6) DNS query types
- This is suspicious, especially since the source IP is IPv4

2. Unusually Long Connection
- Maximum connection duration: 9.420791 seconds
- Could indicate persistent communication

3. Internal Host Identified
- IP: 10.20.57.3
- Initiated excessive DNS queries, all to the same external domain

4. True Positive Confirmed
- Volume + Pattern + Destination consistency confirms DNS tunneling behavior

Indicators of Compromise (IOCs)

TYPE	VALUE
SOURCE IP	10.20.57.3
QUERY TYPE	AAAA (320 instances)
LONGEST CONNECTION	9.420791s
PATTERN	Repetitive DNS queries to the same domain

Analysis Summary

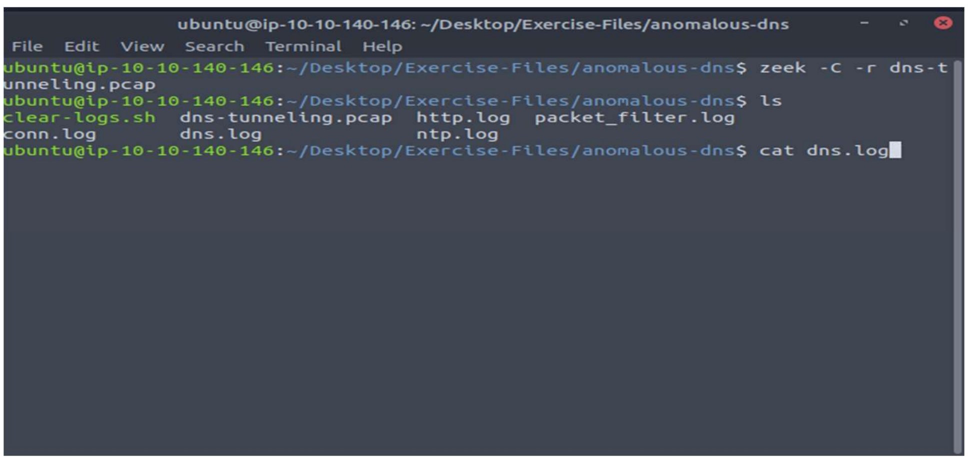
Zeek was effective in uncovering suspicious DNS activity indicative of DNS tunneling, a known exfiltration method. The consistent use of AAAA queries by a single internal host, along with the prolonged connection durations, raises a red flag.

Recommendations

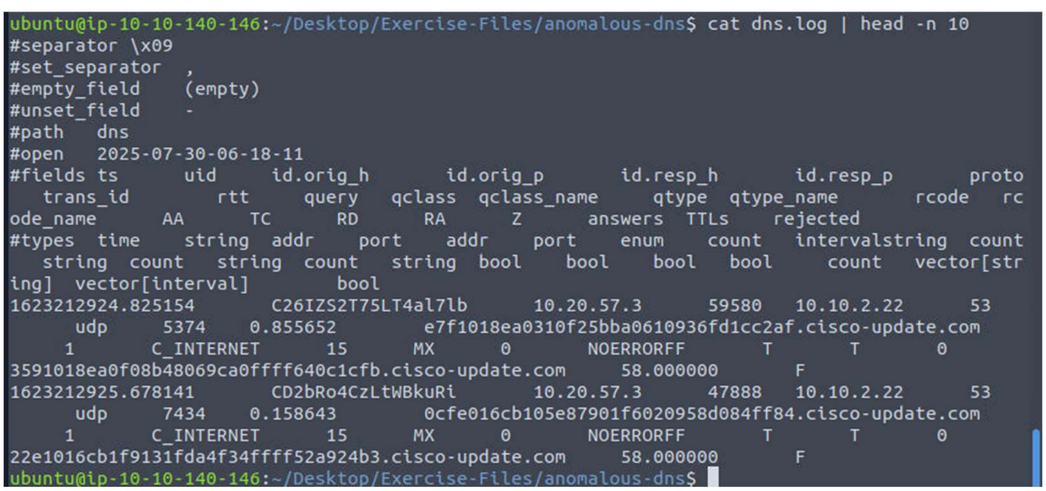
- Monitor all future AAAA query spikes from internal IPv4-only hosts
- Implement deep packet inspection or DNS proxy filtering
- Consider adding thresholds or alerts in SIEM based on similar patterns

Screenshots

- Screenshot 1: Zeek command run on PCAP



- Screenshot 2: Output of dns.log



- Screenshot 3: AAAA count using zeekcut

```
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$ cat dns.log | zee
k-cut qtype_name | sort | uniq -c
     11 A
    320 AAAA
   2232 CNAME
   2314 MX
     23 PTR
   2347 TXT
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$
```

- Screenshot 4: Longest connection from conn.log

```
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$ cat conn.log | he
ad -n 10
#separator \x09
#set_separator  ,
#empty_field    (empty)
#unset_field    -
#path   conn
#open   2025-07-30-06-33-19
#fields ts      uid     id.orig_h       id.orig_p       id.resp_h       id.resp_
p       proto   service duration        orig_bytes      resp_bytes      conn_sta
te      local_orig      local_resp      missed_bytes    history orig_pkts       o
rig_ip_bytes    resp_pkts       resp_ip_bytes   tunnel_parents
#types  time    string  addr    port    addr    port    enum    string  interval
count   count   string  bool    bool    count   string  count   count   count   c
ount    set[string]
1623212924.825154       CH3n6F3H8DzwdXOsmi      10.20.57.3      59580   10.10.2.
22      53      udp     dns     0.855652        80      175     SF      -       -
0       Dd      1       108     1       203     -
1623212925.678141       CTzNeiZTb8CNmmwXh       10.20.57.3      47888   10.10.2.
22      53      udp     dns     0.158643        80      175     SF      -       -
0       Dd      1       108     1       203     -
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$
```

```
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$ cat conn.log | ze
ek-cut duration | sort | uniq
```

- Screenshot 5: Internal IP output

```
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$ cat dns.log | zee
kcut query
```

```
ubuntu@ip-10-10-53-244: ~/Desktop/Exercise-Files/anomalous-dns
File  Edit  View  Search  Terminal  Help
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
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10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
10.20.57.3      udp     dns
```